

By 2nd De Morgan's Law, $\exists x(\neg(\neg A) \wedge \neg P(x))$ is True. By Double-negation law, $\exists x(A \wedge \neg P(x))$ is True. By defn. of existential quantifier, there exists at least one element a in the domain such that $A \wedge \neg P(a)$ is True.

By defn. of AND, A is True and $\neg P(a)$ is True. By defn. of negation, $\neg(\neg P(a))$ is False. By Double negation law, $P(a)$ is False. By defn. of existential quantifier, $\exists x P(x)$ is False, for at least one such element a .

By defn. of negation $\neg \exists x P(x)$ is True. By De Morgan's law of quantifiers, $\forall x(\neg P(x))$ is True. By defn. of negation, $\forall x \neg(\neg P(x))$ is False. By Double-negation law, $\forall x P(x)$ is False. By defn. of conditional, $A \rightarrow \forall x P(x)$ is False.

$\therefore \forall x(A \rightarrow P(x))$ is False $\Rightarrow A \rightarrow \forall x P(x)$ is False

Let, $A \rightarrow \forall x P(x)$ be False. By defn. of conditional, A is True and $\forall x P(x)$ is False. From the defn. of universal quantifier, there exists an element a in the domain s.t. $P(a)$ is False. By the defn. of conditional, $A \rightarrow P(a)$ is False. From the defn. of universal quantifier, $\forall x(A \rightarrow P(x))$ is False. $\therefore A \rightarrow \forall x P(x)$ is False $\Rightarrow \forall x(A \rightarrow P(x))$ is False

$\therefore$ They are logically equivalent $\square$

b) $\exists x(A \rightarrow P(x)) \equiv A \rightarrow \exists x P(x)$

Let, $\exists x(A \rightarrow P(x))$ be False. By defn. of negation, $\neg \exists x(A \rightarrow P(x))$ is True. By De Morgan's law of quantifiers, $\forall x(\neg(A \rightarrow P(x)))$ is True. By conditional disjunction equivalence, $\forall x \neg(\neg A \vee P(x))$ is True. By 2nd De Morgan's law, $\forall x(\neg(\neg A) \wedge \neg P(x))$ is True. By double-negation law, $\forall x(A \wedge \neg P(x))$ is True. From the defn. of universal quantifier, for all elements a in the domain, $A \wedge \neg P(a)$ is True. By the defn. of AND, A is True and $\neg P(a)$ is True. By the defn. of negation, $\neg(\neg P(a))$ is False. By the double-negation law, $P(a)$ is False for all elements a in the domain.